

Data Structures & Algorithms - Core Patterns & Cheatsheet

Curated study guide for technical coding interviews (LeetCode / FAANG)

1. Two Pointers Pattern

Usage: Sorted arrays, searching pairs, palindromes, container with most water.

Key Idea: Left and right pointers moving towards each other or at varying speeds.

Time Complexity: $O(N)$, Space Complexity: $O(1)$.

Classic Problems: 3Sum, Valid Palindrome, Two Sum II, Trapping Rain Water.

2. Sliding Window Pattern

Usage: Subarrays/substrings meeting criteria (maximum sum, longest without repeats).

Fixed Window: Advance right, subtract left once size exceeds k .

Dynamic Window: Expand right to find valid window; shrink left until condition holds.

Classic Problems: Longest Substring Without Repeating Characters, Min Window Substring.

3. Fast and Slow Pointers (Floyd Cycle Detection)

Usage: Linked list cycle detection, finding middle of linked list, happy number.

Key Idea: Slow advances 1 step, fast advances 2 steps. They meet if a loop exists.

To find start of cycle: Reset slow to head, advance both by 1 step until they meet.

4. Dynamic Programming Core Framework

Step 1: Define state (what do indices $dp[i][j]$ represent?).

Step 2: Find base cases (smallest trivial subproblems).

Step 3: State transition recurrence (decision tree choices).

Step 4: Space optimization (e.g., reduce $O(N^2)$ space to $O(N)$ or $O(1)$ if only last row needed).